

Assignments of preliminary introduction to matrix analysis in Cha 1

1. Let the square matrix of order p be

$$A = \begin{pmatrix} 1 & \rho & \cdots & \rho \\ \rho & 1 & \cdots & \rho \\ \vdots & \vdots & \ddots & \vdots \\ \rho & \rho & \cdots & 1 \end{pmatrix}$$

Try to prove that

- (1) $|A| = (1 - \rho)^{p-1}[1 + (p - 1)\rho]$;
- (2) A is the degenerate matrix if and only if $\rho = 1$ or $\rho = -1/(p - 1)$;
- (3) Among the p eigenvalues of A , there are $p - 1$ eigenvalues of $1 - \rho$ and one eigenvalue of $1 + (p - 1)\rho$.

2. Assume A and B both are the $n \times n$ squared matrix. For any $\lambda \neq 0$, prove that

$$|\lambda I_n - AB| = |\lambda I_n - BA|.$$

3. The quadratic form $\sum_{i=1}^n (x_i - \bar{x})^2$ is expressed as $\mathbf{x}'A\mathbf{x}$, where $\mathbf{x} = (x_1, x_2, \dots, x_n)'$, A is a symmetric matrix.

- (1) Find out A ;
- (2) Prove that A is a projection matrix;
- (3) Judge whether A is a positive definite matrix or nonnegative definite matrix;
- (4) Find the rank of A .

4. Let $A = \begin{pmatrix} 9 & -4 & 2 \\ -4 & 3 & 2 \\ 2 & 2 & 10 \end{pmatrix}$, for any nonzero vector $\mathbf{x} = (x_1, x_2, x_3)'$, find the maximum and minimum of $\mathbf{x}'A\mathbf{x}/\mathbf{x}'\mathbf{x}$.

5. Let A be a symmetric matrix. Try to prove that the rank of A is equal to the number of non-zero eigenvalues of A .

6. Obtain a singular value decomposition for the matrix

$$A = \begin{bmatrix} 1 & 2 & 2 & 1 \\ 1 & 1 & 1 & -1 \end{bmatrix}.$$

7. We know that a 3×3 matrix A has eigenvalues $\lambda_1 = 1, \lambda_2 = 2, \lambda_3 = 3$, and corresponding eigenvectors $x_1 = (1, 1, 2)'$, $x_2 = (-1, 1, -2)'$ and $x_3 = (1, 1, 1)'$. Find A .

8. Consider the matrix

$$A = \begin{bmatrix} 0 & -1 & 2 \\ 0 & -1 & 2 \\ 3 & 2 & -1 \end{bmatrix}.$$

a) Find the Morre-Penrose inverse of AA' , and use the property (g) of M-P inverse:

$$A^+ = (A'A)^+A' = A'(AA')^+$$

to find A^+ .

b) Use A^+ to find the projection matrix for the range (=the column space) of A and the projection matrix for the row space of A .

Please submit your assignments before class on Monday of Week 3 (10: 20, Mar 09, 2026).